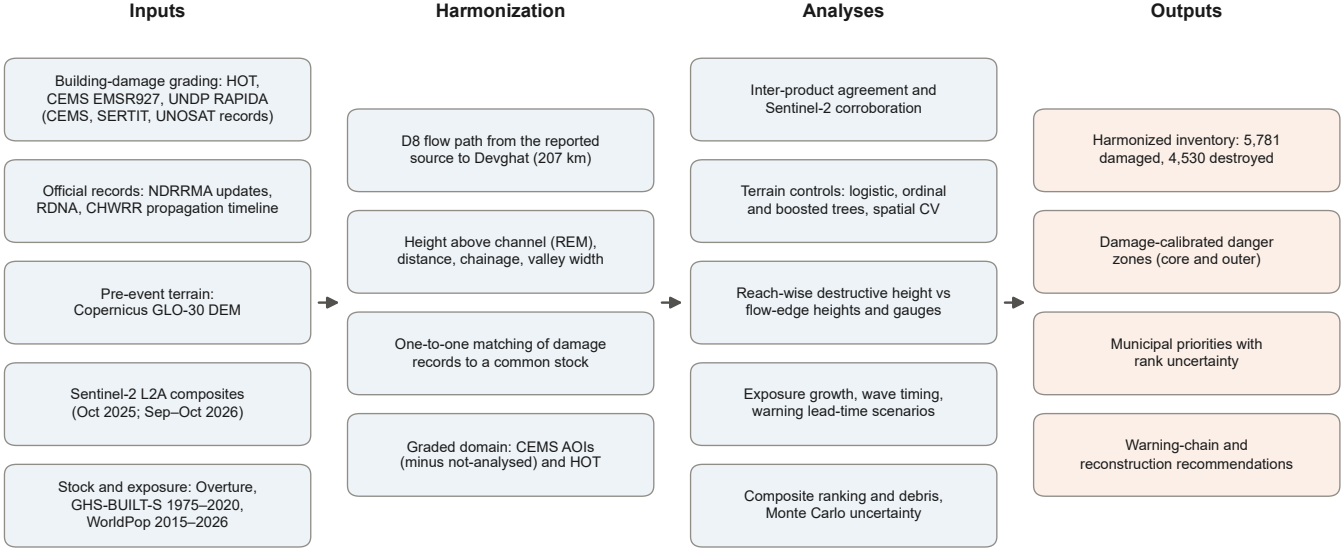
Inputs
Harmonization
Analyses
Outputs
Building-damage grading: HOT, CEMS EMSR927, UNDP RAPIDA (CEMS, SERTIT, UNOSAT records)
Official records: NDRRMA updates, RDNA, CHWRR propagation timeline
Pre-event terrain: Copernicus GLO-30 DEM
Sentinel-2 L2A composites (Oct 2025; Sep–Oct 2026)
Stock and exposure: Overture, GHS-BUILT-S 1975–2020, WorldPop 2015–2026
D8 flow path from the reported source to Devghat (207 km)
Height above channel (REM), distance, chainage, valley width
One-to-one matching of damage records to a common stock
Graded domain: CEMS AOIs (minus not-analysed) and HOT
Inter-product agreement and Sentinel-2 corroboration
Terrain controls: logistic, ordinal and boosted trees, spatial CV
Reach-wise destructive height vs flow-edge heights and gauges
Exposure growth, wave timing, warning lead-time scenarios
Composite ranking and debris, Monte Carlo uncertainty
Harmonized inventory: 5,781 damaged, 4,530 destroyed
Damage-calibrated danger zones (core and outer)
Municipal priorities with rank uncertainty
Warning-chain and reconstruction recommendations